

AI-Driven Electromagnetic Compatibility Assessment in Modern Power Systems

Karuna Pashte Chavande
AI Department
Millennium Techlink Pvt Ltd
Pune MH, India

karuna.pashte@millenniumtechlink.com

Mahima BM
AI Department
Millennium Techlink Pvt Ltd
Pune MH, India

Mahima.BM@millenniumtechlink.com

Dr. Ananth Krishna Dasari
AI and EMC Department
Millennium Techlink Pvt Ltd
Pune MH, India
Ananth.D@millenniumsemi.com

Abstract—The assessment of electromagnetic compatibility (EMC) in power systems has become increasingly complex due to the widespread adoption of renewable energy technologies, advanced power electronics, and smart grid infrastructure. Conventional evaluation techniques often struggle to cope with the dynamic and non-linear characteristics of contemporary electrical networks. This paper investigates the growing application of artificial intelligence (AI)—encompassing machine learning, deep learning, and neural network models—in improving EMC analysis. AI-driven approaches have shown marked advancements in detecting faults, enhancing power quality, and mitigating electromagnetic interference. However, several implementation barriers remain, including issues related to data integrity, computational intensity, and the interpretability of AI models. This work reviews recent developments, presents core methodologies, and proposes future research pathways for AI-enabled EMC assessment in modern power systems.

Keywords—*Electromagnetic Compatibility (EMC), Power Systems, Artificial Intelligence (AI), Machine Learning (ML), Smart Grid, Interference Detection, EMC Mitigation*

I. INTRODUCTION

Modern power systems are becoming increasingly complex due to the integration of renewable energy sources, advanced power electronic devices, and smart grid technologies. These developments have intensified the challenges associated with electromagnetic compatibility (EMC), which ensures that electrical components operate without causing or being affected by electromagnetic interference. Traditional EMC assessment methods, often based on static and linear models, are insufficient for capturing the dynamic and non-linear behavior of today's grids. Artificial intelligence (AI), particularly machine learning (ML), deep learning (DL), and neural networks, has emerged as a promising solution to enhance EMC evaluation. These techniques offer improved accuracy, faster processing, and real-time adaptability. Recent research has focused on combining AI with computational electromagnetics (CEM) to address EMC issues such as fault detection, power quality improvement, and interference mitigation. AI models trained on data from phasor measurement units (PMUs), electromagnetic field sensors, and substation disturbance records have demonstrated superior performance compared to conventional approaches. This paper reviews the current state of AI-driven EMC assessment, explores its integration with smart grid technologies, and discusses theoretical and practical implications. It also highlights key challenges and outlines future directions for research and implementation.

II. LITERATURE REVIEW

A. Evolution of AI in EMC Assessment

The application of artificial intelligence (AI) in the domain of electromagnetic compatibility (EMC) evaluation within power systems has seen a notable surge in recent years. Researchers have increasingly recognized the limitations of traditional EMC assessment techniques, particularly in the context of modern, highly dynamic power networks. As a result, there has been a shift toward leveraging AI technologies to address complex EMC challenges, ranging from fault detection and classification to power quality optimization and electromagnetic interference (EMI) mitigation. One of the key areas of advancement involves the integration of AI algorithms with computational electromagnetics (CEM), enabling more accurate modeling and analysis of electromagnetic phenomena in power system components. This fusion allows for enhanced diagnostic capabilities and predictive maintenance strategies, especially in environments where conventional analytical methods fall short due to system non-linearities and high-frequency switching behaviors.

B. Methodologies and Data Sources

The methodologies employed in current research span a wide spectrum of AI techniques, including both supervised and unsupervised learning models. Supervised learning algorithms such as decision trees, support vector machines (SVM), and ensemble methods are commonly used for fault classification and anomaly detection. Unsupervised approaches, including clustering and dimensionality reduction techniques, are applied to uncover hidden patterns in EMC-related data.

Data used for training and validation of these models is sourced from a combination of simulated environments and actual field measurements. Simulation platforms provide controlled conditions for algorithm development, while real-world data ensures practical relevance and robustness. Key data sources include:

- **Phasor Measurement Units (PMUs):** Offering high-resolution time-synchronized voltage and current data for dynamic system analysis.
- **Electromagnetic Field Sensors:** Capturing spatial and temporal variations in field strength for non-contact fault detection.

- Substation Disturbance Logs: Recording transient events and anomalies for historical pattern recognition and model training.
- These diverse datasets enable AI models to learn from a wide range of operating conditions and disturbances, improving their generalization and adaptability.

C. Key Findings and Performance Outcomes

Studies consistently report that AI-enhanced EMC assessment systems outperform traditional techniques in several critical areas. Notably, machine learning models have demonstrated superior accuracy in fault detection tasks, often achieving classification rates above 95%. Deep learning architectures, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have shown exceptional capabilities in pattern recognition and temporal analysis, making them ideal for EMI source identification and power quality monitoring. In terms of computational efficiency, AI models significantly reduce processing time and resource consumption. For example, EMC diagnostic systems powered by AI have achieved up to threefold improvements in calculation speed and substantial reductions in data storage requirements. These performance gains are crucial for real-time applications where timely decision-making is essential to maintain system stability and reliability. Overall, the literature underscores the transformative potential of AI in EMC assessment, highlighting its ability to enhance diagnostic precision, operational efficiency, and adaptability in complex power system environments.

III. EMC CHALLENGES ADDRESSED BY AI

Artificial intelligence has emerged as a powerful tool for tackling key electromagnetic compatibility (EMC) issues in modern power systems. By leveraging data-driven models and adaptive algorithms, AI enhances the precision, speed, and scalability of EMC diagnostics and control.

A. Power Quality Issues

Power quality disturbances such as harmonics, voltage fluctuations, and resonance can degrade system performance and damage sensitive equipment. AI-based solutions, particularly Unified Power Quality Conditioners (AI-UPQC), have shown remarkable effectiveness in mitigating these issues. These systems dynamically adjust to load variations and significantly reduce total harmonic distortion (THD), while also improving power factor and voltage stability.

B. Fault Detection and Classification

Accurate fault identification is critical for maintaining grid reliability. AI models, especially those using machine learning techniques, enable precise fault classification by analysing electromagnetic field data. This approach allows for non-contact fault detection, offering a safer and more compact alternative to traditional current transformer-based methods. AI-powered protective relays have demonstrated accuracy rates exceeding 99% for both single and double

circuit transmission lines, enhancing fault response and system protection.

C. Electromagnetic Interference (EMI) Management

Electromagnetic interference poses a significant threat to system integrity and communication reliability. AI techniques, including pattern recognition via convolutional neural networks (CNNs), facilitate rapid and accurate EMI source identification. Methods such as Spherical Wave Expansion combined with machine learning have improved diagnostic efficiency by a factor of three and substantially reduced data storage requirements, enabling more scalable and responsive EMI management.

IV. CASE STUDIES AND PRACTICAL APPLICATIONS

The practical deployment of AI-driven EMC solutions across various power system environments has demonstrated measurable improvements in performance, reliability, and adaptability. Below are key examples illustrating the real-world impact of these technologies.

A. Railway Power Systems

In electrified railway networks, maintaining power quality is critical due to fluctuating loads and sensitive onboard equipment. The implementation of AI-based Unified Power Quality Conditioners (AI-UPQC) has led to significant enhancements in system performance. Notably, source current total harmonic distortion (THD) was reduced from 25.16% to 1.12%, while load voltage THD dropped from 6.62% to 2.07%. These systems also respond more rapidly to load variations compared to conventional proportional-integral (PI) controllers, ensuring smoother operation and improved energy efficiency.

B. Transmission Line Protection

Machine learning-based fault detection systems using electromagnetic field data have proven highly effective in transmission line protection. These AI models accurately identify fault types and locations without relying on bulky current transformers, offering a compact and non-intrusive alternative. The approach enhances safety, reduces equipment costs, and supports faster fault isolation, contributing to overall grid resilience.

C. Smart Grid Integration

AI plays a pivotal role in integrating renewable energy sources into smart grids while maintaining EMC compliance. Machine learning algorithms enable real-time monitoring and adaptive control of grid stability, addressing the variability and unpredictability introduced by distributed energy resources. This capability supports seamless energy balancing, proactive fault management, and improved coordination across grid components.

V. PERFORMANCE EVALUATION

The integration of artificial intelligence into EMC assessment frameworks has yielded notable improvements in both quantitative metrics and operational responsiveness. This section highlights key performance gains observed across various implementations.

A. Accuracy Enhancements

AI-driven systems have significantly elevated the precision of EMC diagnostics. For instance, the accuracy of EMC identification models has risen from 77.7% to 99.5% using advanced machine learning algorithms. Similarly, fault detection systems powered by AI have achieved recall rates exceeding 97% when analyzing real-world disturbance data, ensuring reliable classification and timely intervention.

B. Computational Efficiency

AI techniques offer substantial gains in processing speed and resource optimization. EMC diagnostic platforms utilizing AI have demonstrated a threefold increase in computational efficiency. Additionally, data storage requirements have been dramatically reduced—by 76% for relational data and 60.7% for attribute sets—making these systems more scalable and suitable for large-scale deployment.

C. Response Time Improvements

Enhanced responsiveness is another critical benefit of AI-based EMC solutions. Power quality conditioners equipped with AI algorithms adapt more quickly to changing load conditions compared to traditional controllers. Real-time analysis capabilities enable proactive system management, allowing operators to detect and mitigate issues before they escalate, thereby improving overall grid stability and reliability.

VI. IMPLICATIONS AND INTEGRATION

A. Practical Relevance

AI-enhanced EMC assessment contributes directly to improved system reliability, reduced downtime, and more efficient maintenance strategies. Its ability to support real-time monitoring and predictive diagnostics makes it a valuable asset in modern power system operations.

B. Theoretical Implications

This research advances the theoretical understanding of AI applications in electrical engineering, particularly in modeling and managing complex electromagnetic phenomena. It lays the groundwork for future innovations in autonomous and intelligent power system management.

C. Integration with Smart Grid Technologies

AI-driven EMC solutions align closely with the goals of smart grid development. They enable enhanced situational awareness, automated fault response, and adaptive control mechanisms, supporting the transition to more resilient and intelligent energy infrastructures.

VII. FUTURE RESEARCH DIRECTIONS

To fully realize the potential of AI in electromagnetic compatibility (EMC) assessment, future research must address current limitations and explore emerging technologies that enhance robustness, scalability, and trustworthiness. Several promising avenues are outlined below.

A. Explainable AI and Trustworthy Systems

Improving the interpretability of AI models is essential for building trust in critical power system applications. Future work should focus on explainable AI (XAI) frameworks that provide transparent decision-making processes, helping operators understand model outputs and ensuring regulatory compliance.

B. Lightweight AI Models for Edge Devices

As power systems increasingly rely on distributed monitoring and control, there is a growing need for AI algorithms optimized for edge computing environments. Research should aim to develop lightweight models that maintain high performance while operating on resource-constrained hardware, enabling real-time EMC diagnostics at the grid edge.

C. Integration with IoT and Cyber-Physical Systems

The convergence of AI with Internet of Things (IoT) technologies and cyber-physical systems presents new opportunities for intelligent EMC management. Future studies should explore seamless integration strategies that leverage sensor networks, real-time data streams, and adaptive control mechanisms to enhance situational awareness and system resilience.

D. Field Validation and Real-World Testing

While simulation-based studies offer valuable insights, real-world validation is critical for assessing the practical viability of AI-driven EMC solutions. Future research should prioritize large-scale field trials across diverse power system configurations to evaluate performance, reliability, and scalability under operational conditions.

VIII. Conclusion

This paper has explored the transformative role of artificial intelligence in enhancing electromagnetic compatibility (EMC) assessment within modern power systems. AI techniques—including machine learning, deep learning, and advanced neural models—have demonstrated significant improvements in fault detection, power quality control, and electromagnetic interference mitigation. These approaches offer superior accuracy, faster response times, and greater adaptability compared to traditional methods. Despite these advancements, challenges remain in terms of data quality, computational demands, and regulatory compliance. Future research should focus on developing interpretable AI models, optimizing algorithms for edge deployment, integrating with IoT and cyber-physical systems, and validating solutions through real-world testing. Overall, AI presents a promising path forward for intelligent, scalable, and resilient EMC management, aligning with the evolving needs of smart grid infrastructures and next-generation power systems.

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